THE COTANGENT COMPLEX OF A SMOOTH RING MAP IS CONCENTRATED IN DEGREE ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof reduces through locality and the standard smooth presentation, naming each input.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of H^0 of the cotangent complex with the module of differentials. Lemma lemma-identify-H0 (583-586, produced as L44135), used at proof lines 1938-1939. Gives the agreement in degree zero.
- Locality of the cotangent complex under localizing the algebra. Lemma lemma-localize-on-top, used at proof lines 1941-1948. Reduces to the standard smooth case and identifies the complex after base change.
- The standard smooth presentation of a smooth ring map. Algebra, Lemma algebra-lemma-smooth-syntomic, used at proof lines 1943-1947. Produces the factorization with an etale second map.
- The vanishing statement and the computation for polynomial algebras. Lemmas lemma-when-zero and lemma-cotangent-complex-polynomial-algebra, used at proof lines 1947-1951. Give the conclusion in the standard smooth case.

Source reference. Original lemma 00TA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1068–1068. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11060–11060. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11058–11058. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08R3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11081–11081. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08R2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11080–11080. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *If $A \to B$ is a smooth ring map, then $L_{B/A} = \Omega_{B/A}[0]$.*

Proof. We have the agreement in cohomological degree 0 by Lemma 08QF. Thus it suffices to prove the other cohomology groups are zero. It suffices to prove this locally on $\mathrm{Spec}(B)$ as $L_{B_g/A} = (L_{B/A})_g$ for $g \in B$ by Lemma 08R3. Thus we may assume that $A \to B$ is standard smooth (Algebra, Lemma 00TA), i.e., that we can factor $A \to B$ as $A \to A[x_1, \ldots, x_n] \to B$ with $A[x_1, \ldots, x_n] \to B$ étale. In this case Lemmas 08R2 and Lemma 08R3 show that $L_{B/A} = L_{A[x_1,\ldots,x_n]/A} \otimes B$ whence the conclusion by Lemma 08QH. □